

Linux Terminal To First Script

Today we build the survival kit: terminal orientation, files, permissions, and a first Python script.

Students will practice

- opening a terminal
- finding the current folder
- creating and viewing files
- reading permissions
- running Python from Linux

Lab target

- most of class is hands-on
- evidence goes to Brightspace
- writing explains what happened
- broken environments use honest documentation

What Are We Actually Using?

The words sound similar, but each layer has a job.

Python script

- > shell command

- > terminal window

- > shell, usually Bash

- > Linux distribution, such as Fedora

- > Linux kernel

- > hardware or virtual machine

When students get lost, point back to this stack.

Linux Shows Up In Real Technical Work

This course is not only about operating-system trivia.

Where Linux appears

- cloud servers
- Docker containers
- backend APIs
- cybersecurity labs
- QA automation
- data and AI workflows

Beginner skill translation

- terminal confidence
- file and path awareness
- script execution
- reading errors
- saving useful output
- explaining technical steps

When You Are Lost, Ask The System

These commands are safe. They help you orient before changing anything.

```
$ whoami
```

```
$ hostname
```

```
$ date
```

```
$ pwd
```

```
$ ls
```

```
$ ls -la
```

Repeat this habit: pwd first, then ls.

Linux Is A Tree Of Files And Folders

The root folder is slash. Your work usually belongs inside your home folder.

```
/
|-- home/
|   |-- student/
|       |-- bootcamp/
|-- etc/      configuration files
|-- var/      logs and changing data
|-- usr/      installed programs
`-- tmp/      temporary files
```

Students should work in ~/bootcamp, not random system folders.

Absolute Paths Vs Relative Paths

A path is an address for a file or folder.

Absolute path

- starts at slash
- works from anywhere
- example: /home/student
- example: /etc/hosts

Relative path

- starts from current folder
- depends on pwd
- example: notes.txt
- example: ../class01

Create, Edit, Inspect

This is the beginner Linux loop.

```
$ mkdir -p ~/bootcamp/class01
```

```
$ cd ~/bootcamp/class01
```

```
$ touch notes.txt
```

```
$ nano notes.txt
```

```
$ cat notes.txt
```

```
$ ls -l notes.txt
```

Nano save: Ctrl+O, Enter, Ctrl+X.

Read, Write, Execute

Permissions answer what a user can do with a file.

```
$ ls -l  
-rw-r--r-- notes.txt  
-rwxr-xr-x script.py
```

r = read
w = write
x = execute

Today: understand the idea, not every numeric permission.

Run A First Script

This connects terminal skills to software development.

```
$ python3 --version
```

```
$ nano hello.py
```

```
print("Hello from Linux")
```

```
print("I can run Python from the terminal")
```

```
$ python3 hello.py
```

The terminal is how many developers run and test code.

Collect Basic System Information

A script can turn commands and facts into reusable output.

```
import os
import platform
import shutil

print("System:", platform.system())
print("Machine:", platform.machine())
print("Python:", platform.python_version())
print("Current folder:", os.getcwd())
```

Students run the full version in the lab.

Errors Are Evidence

Do not guess randomly. Read the message and check location, spelling, and command.

Common causes

- wrong folder
- misspelled filename
- wrong capitalization
- file not saved
- Python not installed
- copied smart quotes

Debug habit

- run pwd
- run ls
- check the exact filename
- read the first error line
- read the last error line
- try one fix at a time

Brightspace Evidence

The lab is not just commands. It includes proof and explanation.

Submit evidence

- pwd output
- ls -la output
- python3 --version output
- hello.py output
- system_info.py output

Submit writing

- what mkdir -p does
- what cat does
- what permissions you saw
- what system info reported
- what still feels confusing

What Success Looks Like Today

No one needs to be fast yet. They need to be oriented.

Students should be able to say

- I can find my current folder
- I can list files
- I can create and view a file
- I can recognize permissions
- I can run a Python script

Next class

- Bash scripts
- pipes
- grep
- find
- text analysis